

Prof. Dr. Lincy Pyl and Selection Committee
Department of Mechanics of Materials and Constructions (MeMC)
Vrije Universiteit Brussel

Subject: Motivation letter – PhD candidate, Mechanics of Materials and Constructions (M²S)
Introduction video: <https://drive.google.com/file/d/14s-V07z3fopEllnUP8vmMJPMxQW57mg9/view>

Dear Prof. Pyl and Selection Committee,

I am Pooja Awasare, a civil and environmental engineer. I am applying for the PhD position in Mechanics of Materials and Constructions at VUB because I want to return to materials and structures at research level, in a group where testing and clear engineering questions come first.

My background starts with a Diploma and a Bachelor's in Civil Engineering in India. In those years I spent a lot of time in the materials lab with concrete, steel, soil, aggregates and water. That work stayed with me. When a result looked wrong, I did not like to leave it. I wanted to understand what caused it. I work in that same way today: I stay with a question and go deeper until the reason is clear. I am consistent, and I am curious enough to keep reading and checking until things make sense.

I later completed a Master's in Environmental Engineering at HTW Dresden. My thesis looked at how construction teams can manage projects and use resources more carefully. After that, one idea became clearer for me: sustainable construction cannot skip materials and structures. They are the heart of building. If we do not understand how materials behave, and how structures use them, then talk about sustainability stays weak.

I have lived and worked in Belgium, including with a construction partner on circular practices in day-to-day project delivery. That experience showed me how industry thinks about waste, reuse and process. It also showed me what I still want to learn: the mechanical side of materials and structures, with proper testing and analysis. That is why I am looking for a PhD now, and why I want to return to Brussels for it.

I have been following MeMC and reading about your group's work. I read about Prof. Pyl's research on lightweight structures and on the mechanical performance of materials, with experiments linked to analytical and numerical models. I also followed MeMC's work on resource-efficient and circular building systems. What attracts me is the seriousness of the questions: how materials perform, how structures can use less material without losing strength, and how careful testing supports better design. That depth matches the way I like to learn.

I know I am still early in a research career. I am willing to prepare applications for external funding such as FWO, and I am ready to learn what the group needs from a PhD student, including teaching support where asked.

I would be glad to discuss a PhD topic with you on the mechanical behaviour of construction materials and more careful structural design for sustainability. Thank you for considering my application and for watching my introduction video.

Yours sincerely,

Pooja Awasare